Task 3: Work with YourTTS

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Abstract

YourTTS, which supports only English, French and Brazilian Portuguese, was used to synthesize an English and a Lithuanian sentence in my voice without any training (zero-shot). The English result was [clear and similar to my voice]; the Lithuanian one, read with the English setting, kept my voice but had [poor] pronunciation. Both had a narrower spectrum (up to 8 kHz) and [flatter] intonation than natural speech. A dataset of [ ] recordings ([ ] min) of EMO speaker [F3], emotion [neutral], was prepared for future fine-tuning.

1. Model and supported languages

YourTTS is a multilingual, multi-speaker text-to-speech model. It can do **zero-shot voice cloning**: from a few seconds of a new speaker's voice it produces speech in that voice, without any extra training. The model was loaded with the **Coqui TTS** library (`tts_models/multilingual/multi-dataset/your_tts`) in Python 3.10.

Supported languages. The command `print(tts.languages)` returned three languages (Table 1). **Lithuanian is not supported**, so a Lithuanian text has to be given one of these three language settings.

Code	Language
en	English
fr-fr	French (France)
pt-br	Portuguese (Brazil)

Table 1. Languages supported by the pretrained YourTTS model.

2. Recordings and analysis method

Reference recordings. I recorded two sentences on an iPhone: the English *"This is a test of the speech synthesis system."* (`reference_en.wav`) and the Lithuanian *"Tai yra šnekos sintezės sistemos testas."* (`reference_lt.wav`). Each recording served both as the **natural speech** sample and as the **voice reference** for YourTTS. The original files were

stereo; before use, the click at the end of each file (the tap that stopped the recording) was cut off and both files were converted to mono. Table 2 lists the parameters of all four signals.

Recording	Duration	Sampling rate	Format	Channels
reference_en.wav (natural)	3.50 s	48 kHz	WAV, 16-bit PCM	1 (mono)
output_en.wav (synthesized)	[ ] s	16 kHz	WAV, [16]-bit	1
reference_lt.wav (natural)	3.60 s	48 kHz	WAV, 16-bit PCM	1 (mono)
output_lt.wav (synthesized)	[ ] s	16 kHz	WAV, [16]-bit	1

Table 2. Parameters of the natural and synthesized recordings.

Analysis. Each signal was loaded with librosa at its original sampling rate. The **waveform** shows amplitude over time; the **magnitude spectrum** (FFT of the whole signal, in dB) shows how energy is spread over frequencies; the **F0 contour** (pYIN algorithm, 60–400 Hz) shows the pitch over time, with no values in unvoiced parts.

3. English zero-shot synthesis

Procedure. The English sentence was synthesized with `tts.tts_to_file(text=..., speaker_wav="reference_en.wav", language="en")`, which produced `output_en.wav` in my voice.

Quality. [2–3 sentences in your own words: Is every word clear? Does it sound like you? Does it sound natural or robotic?]

[Insert plots_en.png here]

Figure 1. English: waveform, magnitude spectrum and F0 contour of natural (left) and synthesized (right) speech.

	EN natural	EN synthesized	LT natural	LT synthesized
Highest frequency	~13 kHz	8 kHz	~13 kHz	8 kHz
Mean F0	123 Hz	[ ] Hz	117 Hz	[ ] Hz
F0 range	97–166 Hz	[ ]–[ ] Hz	60–149 Hz	[ ]–[ ] Hz

Table 3. Spectrum and F0 of natural and synthesized speech (durations are in Table 2).

Duration. The synthesized sentence is [shorter / longer] than mine ([ ] s vs 3.50 s), partly because my recording has silence at the start and end, and partly because the model chooses its own speaking rate.

Spectrum. YourTTS outputs **16 kHz** audio, so its spectrum ends at **8 kHz**, while my recording (48 kHz) has speech energy up to about 13 kHz. The /s/ in *"this"* and *"system"* loses part of its energy, and the voice sounds slightly duller.

F0. The mean F0 is [close to / higher than / lower than] mine (123 Hz) and the F0 range is [smaller], so the intonation is [flatter], which is typical of TTS.

4. Lithuanian zero-shot synthesis

Procedure. Lithuanian is not supported, so the Lithuanian sentence was synthesized with the **English setting** (`language="en"`) and `reference_lt.wav` as the voice reference, giving `output_lt.wav` .

Quality. [2–3 sentences in your own words. Typically the voice still sounds like you, but the text is read with English pronunciation, and letters like "š" and "ė" are skipped or said wrongly (look for "Character … not found" warnings in the console).]

[Insert plots_lt.png here]

Figure 2. Lithuanian: waveform, magnitude spectrum and F0 contour of natural (left) and synthesized (right) speech.

Comparison. The synthesized sentence lasts [ ] s against 3.60 s for my recording (Table 2) [; it is shorter because some letters were dropped]. As in English, the spectrum ends at 8 kHz, which is audible on the fricatives /š/, /s/ and /z/. The F0 contour (Table 3) follows [an English-like / a flat] pattern, not natural Lithuanian intonation.

English vs Lithuanian. The voice is copied about equally well in both languages, but the Lithuanian result is [much less intelligible]: YourTTS can clone a voice in any language, but pronounces well only the languages it was trained on.

5. Lithuanian dataset preparation

The dataset was built from the **Lithuanian Emotional Speech Corpus (EMO)**. Its metadata lists **349,696** transcribed recordings by five speakers (F2, F3, F4 female; M3, M4 male) in five emotions: D – joy, L – sadness, N – surprise, P – anger and Z – neutral. Only **one archive part**, [emo-wavs.zip.009], was downloaded.

Selection. Speaker and emotion are coded in each file name: `0301-F3Z-0001.wav` is speaker **F3**, emotion **Z**. I selected all recordings of speaker **[F3]** in emotion **[Z – neutral]** found in the archive part, because [it was the largest group / neutral speech is easiest to

start with]. The transcriptions were taken from `metadata_of_EMO_dataset.csv`.

Split and format. The files were renamed `0001.wav`, `0002.wav`, …, shuffled (seed 42) and split into **90 % training** and **10 % validation**. The metadata files use the Coqui *ljspeech* format, one line per recording: `0001|transcription|transcription`.

Parameter	Value	Parameter	Value
Speaker	[F3]	Recordings	[ ]
Emotion	[Z – neutral]	Total duration	[ ] min
Sampling rate	[ ] Hz	Minimum duration	[ ] s
Bit depth	[16] bit	Maximum duration	[ ] s
Channels	[1 (mono)]	Average duration	[ ] s
Training set	[ ]	Validation set	[ ]

Table 4. Selected EMO recordings.

[Insert duration_histogram.png here]

Figure 3. Histogram of recording durations for speaker [F3], emotion [Z].

Interpretation. Most recordings last between [ ] and [ ] s, [with a few longer sentences up to [ ] s]. Very short or very long recordings are less useful for training and could be removed before fine-tuning.

6. Conclusions

1. **Languages:** the pretrained YourTTS supports English, French and Brazilian Portuguese; Lithuanian is not supported.
2. **English:** zero-shot synthesis gave [clear] speech in a voice [similar] to mine, but with a narrower spectrum (up to 8 kHz) and [flatter] intonation than natural speech.
3. **Lithuanian:** with the English setting the voice was still copied, but the text was read with English pronunciation and letters like "š" and "ė" were [lost], so zero-shot synthesis is not enough for an unsupported language.
4. **Dataset:** for fine-tuning, [ ] recordings ([ ] min) of speaker [F3], emotion [Z], were prepared from the EMO corpus and split into [ ] training and [ ] validation recordings.